WORKSHEET 1.4

Limit Laws and Direct Substitution

Unit 1: Limits and Continuity

Key ideas

- If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$ are real numbers, limits of sums, differences, products and constant multiples are found termwise.
- The quotient law requires $M \neq 0$.
- For polynomial, rational, root, trigonometric, exponential and logarithmic functions, $\lim_{x\to c} f(x) = f(c)$ at every c in the domain.
- $\lim_{x\to c} f(g(x)) = f\big(\lim_{x\to c} g(x)\big)$ only when f is continuous at that inner limit.

Worked examples

Example 1 · Direct substitution

Evaluate $\displaystyle\lim_{x\to -1} \frac{2x^3 - x + 4}{x^2 + 3}$.

Solution. The denominator at $x = -1$ is $4 \neq 0$, so substitute: $\dfrac{-2 + 1 + 4}{4}$.

The limit is $\dfrac{3}{4}$.

Example 2 · Laws with unknown functions

Given $\lim_{x\to 1} f(x) = 3$ and $\lim_{x\to 1} g(x) = -4$, find $\displaystyle\lim_{x\to 1} \frac{[f(x)]^2 - 2g(x)}{f(x) + g(x) + 5}$.

Solution. Numerator: $9 + 8 = 17$. Denominator: $3 - 4 + 5 = 4 \neq 0$.

The limit is $\dfrac{17}{4}$.

Example 3 · Working backwards

Suppose $\displaystyle\lim_{x\to 5} \frac{f(x) - 2}{x - 5} = 7$. Find $\lim_{x\to 5} f(x)$.

Solution. $f(x) - 2 = \dfrac{f(x) - 2}{x - 5} \cdot (x - 5) \to 7 \cdot 0 = 0$.

$\lim_{x\to 5} f(x) = 2$.

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Practice

CORE

1. Evaluate $\lim_{x\to 2}\left(x^4 - 3x\right)$.

2. Evaluate $\lim_{x\to \pi/4} \sin x \cos x$.

3. Evaluate $\lim_{x\to 0} \dfrac{3e^x - 2}{x+1}$.

4. Evaluate $\lim_{x\to 4} \sqrt{x^2+9}$.

EXAM LEVEL

5. Given $\lim_{x\to 3} f(x) = -2$ and $\lim_{x\to 3} g(x) = 6$, find $\lim_{x\to 3} \left[g(x) - (f(x))^2\right] f(x)$.

6. With the same f and g, find $\lim_{x\to 3} \sqrt{g(x) + f(x) + 5}$.

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7. With the same f and g, what can be said about $\lim_{x\to 3} \dfrac{f(x)}{g(x)-6}$? Explain.

8. Suppose $\lim_{x\to -1} \dfrac{f(x)+4}{x+1} = 3$. Find $\lim_{x\to -1} f(x)$.

Question 9 · Free-response practice

Let $f(x) = 4 + (x-2)^2$ and $g(u) = \begin{cases} u^2, & u \le 4 \\ 10-u, & u > 4. \end{cases}$

(a) Find $\lim_{x\to 2} f(x)$, and state whether $f(x)$ approaches this value from above, from below, or from both sides.

(b) Explain why $\lim_{x\to 2} g(f(x))$ cannot be found by computing $g(\lim_{x\to 2} f(x))$.

(c) Find $\lim_{x\to 2} g(f(x))$.

(d) Find $\lim_{x\to 2} g(k(x))$ where $k(x) = 4-(x-2)^2$.

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Worked solutions

1. $16 - 6 = 10$.
2. $\dfrac{\sqrt{2}}{2} \cdot \dfrac{\sqrt{2}}{2} = \dfrac{1}{2}$.
3. $\dfrac{3-2}{1} = 1$.
4. $\sqrt{25} = 5$.
5. $(6-4)(-2) = -4$.
6. $\sqrt{6-2+5} = 3$.
7. It does not exist. The denominator tends to 0 while the numerator tends to $-2 \neq 0$, so the quotient is unbounded; the quotient law does not apply.
8. $f(x) + 4 = \dfrac{f(x)+4}{x+1}(x+1) \to 3 \cdot 0 = 0$, so the limit is -4.
9. (a) 4. Since $(x-2)^2 > 0$ for $x \neq 2$, $f(x) > 4$: it approaches 4 from above only.
   (b) g is not continuous at $u = 4$: $\lim\limits_{u \to 4^-} g(u) = 16$ but $\lim\limits_{u \to 4^+} g(u) = 6$.
   (c) $f(x) \to 4^+$, so the limit is $\lim\limits_{u \to 4^+} g(u) = 6$.
   (d) Now $k(x) < 4$ for $x \neq 2$, so $k(x) \to 4^-$ and the limit is 16.

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